

Key Skills: PLC / TIA Portal • ROS2 / Gazebo • Quality Control & Root Cause Analysis • Profinet/Modbus • Computer Vision • Factory IO

Professional Profile

Final-year (5th year) Industrial Computing and Automation student at INSAT, with hands-on experience across PLC programming, robotics, and aircraft-system quality control.

At Lacroix Electronics, I helped trace a robot arm malfunction back to a faulty sensor feeding bad data to the PLC – the kind of automation-meets-quality troubleshooting I want to keep doing.

Looking for a final-year internship in automation, robotics, or quality engineering, starting February 2027.

Education

Engineering Diploma, Industrial Computing and Automation – Intelligent Systems track2022 – 2027

INSAT – National Institute of Applied Sciences and Technology, Tunis (Machine Learning, Big Data, Deep Learning, Data Science, Operations Research, UML)

Professional Experience

- Intern, Industrial Engineering2024
- Lacroix Electronics, Tunisia
- PLC programming:** programmed Siemens PLCs in TIA Portal and wrote Ladder logic for production sequences; worked with Profinet and Modbus on the plant's OT network.
 - Root cause diagnosis:** helped identify a robot arm malfunction by applying Lean root-cause tools, tracing the issue to a sensor sending faulty signals to the PLC.
 - Broad exposure to quality practices alongside the automation work on the line.
- Intern, Aircraft Systems – Quality Control & Maintenance2025
- Tunisair Technics, Tunis, Tunisia
- Quality control and maintenance of aircraft hydraulic and pneumatic systems**, in a high-reliability aerospace environment with strict traceability and preventive-maintenance requirements.
- Intern, APS Department (Autoliv Production System)Jun – Aug 2025
- Autoliv, El Fahs, Tunisia
- Helped build the plant's **5S Lean dashboard in Python (openpyxl)** and studied the factory layout in AutoCAD as part of a broader equipment-allocation and process-digitalization project.

Technical Projects

Autonomous Industrial Inspection Robot – ROS2 Humble, Gazebo Classic, Nav2, ESP32/micro-ROS, PyQt6, Vosk offline speech recognition; SLAM-based mapping and navigation, real-time monitoring dashboard, supervised by Pr. Saber Tlili.

Embedded Industrial Vision System for Quality Control – Raspberry Pi 5, OpenCV, Python; automatic visual sorting, sub-millimetric label-position checking, OCR-based manufacturing-date validation.

Factory IO & TIA Portal Project – simulated a factory floor controlled by a Siemens PLC, with KPI tracking and visualization linked to Excel.

Additional projects: industrial digital twin, SAP production planning, AnyLogic simulation, VoltMind AI assistant, operations research logistics study, project management project, early line-follower robot.

Technical Skills

Automation:	PLC/TIA Portal, Ladder logic, Profinet, Modbus, Factory IO
Robotics:	ROS2, Gazebo, Nav2, SLAM, ESP32/micro-ROS
Quality & Root Cause:	Root cause analysis, 5 Why, 8D, Ishikawa, AMDEC, 5S, aircraft hydraulic/pneumatic systems QC
Vision & Data:	Python (OpenCV), Computer Vision, Excel (VBA), Power BI
Languages:	Arabic (native), French (C1), English (C1)

Leadership & Distinctions

Chairwoman, IEEE Women in Engineering – INSAT SAG (2024–2025) – led a **100+ member** organization: budget, volunteer team, and end-to-end technical events.

Team Lead, VIP & Protocol Department – National Robotics Week 8.0 (2026) – ran the VIP & Protocol team and developed panel content (automotive/SDV, cyber-physical security) for a national robotics event.

Robotics Instructor, Play Mind Club (2025 – Jun 2026) – ran STEM workshops (Scratch, Arduino, robotics) for children aged 8–12.